
LECTURE NOTES

Lecture 12: Optimization in Machine Learning

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1 Theoretical Analysis - Preliminaries

We know that for

$$F(w) = \begin{cases} R_n(w) = \frac{1}{n} \sum_{i=1}^n f_i(w) & \text{Emperical Risk} \\ R(w) = \mathbb{E}_{\Xi}[f(w; \Xi)] & \text{Expected Risk} \end{cases}$$

Procedure SG(...)

```
1 | Choose an inital iterate  $w_0$ 
  | for  $k = 0, 1, \dots$  do
2 |   Generate a realization of the random variable  $\Xi_k$ ;
3 |   Compute a stochastic vector  $g(w_k, \Xi; k)$ ;
4 |   Choose a stepsize  $\alpha_k > 0$ ;
5 |   Set the new iterate as  $w_{k+1} \leftarrow w_k - \alpha g(w_k, \Xi_k);$ 
```

2 Convergence Analysis - Assumptions

$$g(w_k, \xi_k) = \begin{cases} \nabla f((w_k; \xi_k)) \\ \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f((w_k; \xi_k)) \\ H_k \cdot \frac{1}{n_k} \sum_{i=1}^{n_k} \nabla f((w_k; \xi_k)) \end{cases}$$

3 Convergence Analysis - Results

4 Computational Complexity Analysis